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To the Editors, *Physical Review B*

Re: Submission of “Algebraic leak and thermal hybridization, not athermality, defeat quantum many-body scar codes under generic measurement”

We submit for your consideration the manuscript above (regular article, two authors, twelve pages including nine appendices) together with an openly available simulation package that regenerates every figure with one command.

Quantum many-body scars—non-thermal, weakly entangled, dynamically coherent eigenstates—are widely regarded as natural candidates for protected quantum memory. We test this expectation directly in the monitored-dynamics setting where information protection is sharply defined: encoding a logical qubit in the canonical PXP scar tower and computing its reference-qubit coherent information under projective monitoring, benchmarked against an energy-matched ensemble of thermal encodings. The scar code loses under every generic measurement we test—local, collective, and coarse-grained Casimir probes, across the full measurement-rate and block-size grid, with hierarchical-bootstrap confidence intervals excluding zero at every headline point—and where information survives, the local deficit *grows* with system size (-0.16 to -0.38 for $L = 10$ to 18).

The contribution beyond the negative result is mechanistic, and it is eigenstate-resolved in a way we believe will interest the PRB readership. Two independent structural liabilities are identified and separated: (i) an algebraic leak—the emergent $\text{su}(2)$ order parameter that generates the scar tower couples the code states, so any measurement aligned with it reads the logical qubit out; and (ii) thermal hybridization—local-measurement protection rises monotonically with the forward-scattering-approximation purity of the rung, crossing over to a genuine scar *advantage* at the purest rungs. Athermality per se is neither necessary nor sufficient: what decides the fate of a weak-ergodicity-breaking code is the interplay of its symmetry algebra with the measurement basis and its residual overlap with the thermal bulk. An exactly solvable spin-1 model with zero Casimir variance anchors the clean limit and correctly predicts the unique fine-tuned probe under which scars are rescued.

Relative to prior work: (i) the first head-to-head coding comparison of scar versus thermal subspaces under monitoring at the level of channel capacity rather than entanglement or fidelity; (ii) a purity-resolved mechanism connecting eigenstate hybridization to operational code performance; (iii) a documented sign-reversing baseline artifact—single-eigenstate references invert the conclusion; only ensemble averaging is safe—of direct relevance to anyone benchmarking codes against individual eigenstates.

Physical Review B is a natural home for this work: quantum many-body scars, the PXP model, and measurement-induced phase transitions are core PRB topics, and the paper’s central objects—eigenstate hybridization, emergent symmetry algebras, and their operational consequences—sit

squarely in the condensed-matter tradition of connecting spectral structure to physical response. The manuscript is fully self-contained (appendices in place of a supplement), the data are archived with a DOI ([10.5281/zenodo.21642056](https://doi.org/10.5281/zenodo.21642056)), and the literature claims have been verified against primary sources.

We suggest the following referees, all leading experts in measurement-induced dynamics, quantum error correction, or quantum many-body scars (institutional contact addresses are publicly listed):

- Prof. Matthew P. A. Fisher, University of California, Santa Barbara — measurement-induced transitions
- Prof. Ehud Altman, University of California, Berkeley — error correction and monitored circuits
- Dr. Michael J. Gullans, NIST / University of Maryland — purification transitions
- Prof. Vedika Khemani, Stanford University — random and monitored circuits
- Prof. Adam Nahum, École Normale Supérieure, Paris — entanglement dynamics
- Prof. Zlatko Papić, University of Leeds — quantum many-body scars
- Prof. Maksym Serbyn, IST Austria — quantum many-body scars
- Dr. Sanjay Moudgalya, Technical University of Munich — exact scars and spectrum-generating algebras

We do not request the exclusion of any referee. This manuscript was previously considered at PRX Quantum, where it was returned at the editorial stage on selectivity grounds without technical criticism; it has since been substantially strengthened through three rounds of adversarial internal review (hierarchical-bootstrap confidence intervals, multi-seed audits, and a size-dependence analysis added).

On behalf of both authors,

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